FIZ 101 · Physics I: Mechanics · Fall 2026 · Week 4

Applying Newton's laws

Lecture notes · Mehmet Özkan, Istanbul Technical University

6 Applying Newton's laws

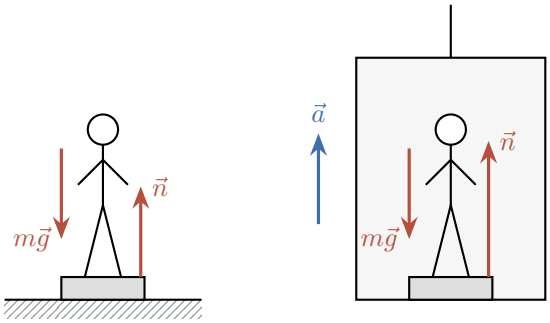
In Chapter 5 we met Newton's three laws of motion. In this chapter we put them to work. We look at the push of a floor and the pull of a rope on a body that accelerates, which leads us to the idea of *apparent weight* (Section 6.1); at friction between dry surfaces (Section 6.2); and at the resistance of a fluid such as air or water to a body moving through it (Section 6.3). We close with the dynamics of circular motion (Section 6.4).

In every problem we follow the recipe of Section 5.4: we isolate each body, draw its free-body diagram with only the forces that act *on* it, choose axes, and write $\sum F_x = ma_x$ and $\sum F_y = ma_y$ in an inertial frame. Two new ingredients appear in this chapter. The first is *constraints*: an unstretchable rope makes the two bodies at its ends move together. The second is a family of forces, such as the push of a floor, the tension of a rope or static friction, that are not given by a formula: they adjust themselves, within limits, to whatever the motion requires, and we find them from Newton's second law. Unless we say otherwise, we take $g = 9.8\,\mathrm{m/s^2}$, neglect air resistance (except in Section 6.3), and treat ropes and strings as massless and unstretchable.

6.1 Apparent weight

Weight is a word we use every day, yet in physics it is one of the most misleading ones, because in everyday language it means at least three different things: the mass of a body ("I weigh 70 kg"), the pull of gravity on it, and the reading of a scale. Recall from Section 5.3 that the **weight** of a body is the gravitational force that the Earth (or the Moon, or whatever planet we are on) exerts on it, $\vec{w} = m\vec{g}$. Its magnitude $w = mg$ is measured in newtons; it depends on where the body is, but not on how the body moves. A bathroom scale, however, does not measure this force directly: it measures how hard the body presses on it. To see what that means, let us look at the example in Figure 6.1.

On the left of Figure 6.1 a man stands on a scale. Two forces act on him: the pull of gravity, mg, downward, and the upward push of the scale on his feet, a normal force $\vec{n}$. He does not accelerate, so by Newton's first law the net force on him is zero, and $n = mg$. Note that $\vec{n}$ and $m\vec{g}$ are *not* an action–reaction pair, even though here they are equal and opposite: both act on the same body, the man, and they are equal only because he does not accelerate. The third-law partner of $\vec{n}$ is the push of the man's feet on the scale, which has the same magnitude n. This is the force that compresses the spring inside the scale, so it is n that the scale reads. (The partner of his weight is the gravitational pull of the man on the Earth.) We call the reading of the scale, that is, the magnitude of the force that the support exerts on the body, the body's **apparent weight**. For a man at rest it is equal



(a) at rest (b) accelerating upward

Figure 6.1. Weight is misleading. (a) A man standing on a scale. Two forces act on him: his weight $m\vec{g}$ and the upward push $\vec{n}$ of the scale. At rest, $n = mg$. (b) The same man on the same scale in an elevator that accelerates upward: now $n = m(g + a) > mg$, and the scale reads more than his weight.

to his weight, mg. If you take the measurement on the surface of the Moon, where the gravitational acceleration is about six times smaller ($1.62\,\mathrm{m/s^2}$), you find your weight six times smaller, although your mass is the same. (A bathroom scale that shows kilograms simply divides the force by $9.8\,\mathrm{m/s^2}$; on the Moon it would show about a sixth of your mass.)

On the right of Figure 6.1 the same man stands on the same scale in an elevator that accelerates upward, in the $+y$ direction, with acceleration a. We apply Newton's second law in the frame of the building, which we treat as inertial (Section 5.2); the accelerating elevator is *not* an inertial frame. The forces on the man are still his weight mg, in the $-y$ direction, and the normal force n of the scale, in the $+y$ direction. Since he accelerates in the $+y$ direction together with the elevator, the net force on him must point up, so clearly $n > mg$. The second law gives

$$ma = n - mg, \tag{6.1}$$

so the scale reads

$$n = m(g + a). \tag{6.2}$$

So if you accelerate in the direction opposite to gravity in an elevator, you look heavier on the scale! Your weight mg has not changed; only the scale reading, your apparent weight, has. In an elevator accelerating upward at $5\,\mathrm{m/s^2}$, the scale reads $(9.8 + 5)/9.8 \approx 1.5$ times what it reads when the elevator is at rest. (Real elevators accelerate at only about $1\,\mathrm{m/s^2}$; we exaggerate to make the effect clear.)

What matters here is the direction of the *acceleration*, not the direction of the velocity. An elevator that is going up and speeding up and one that is going down and slowing down (just before it stops at a lower floor) both have an upward acceleration, and in both the scale reads more than mg. In an elevator moving with constant velocity, up or down, $a = 0$ and the scale reads exactly mg.

If we solve the same problem for an elevator that accelerates in the $-y$ direction, with an acceleration of magnitude a, the net force on the man must point down, and the second law (with the positive direction now taken downward, along the acceleration) gives the

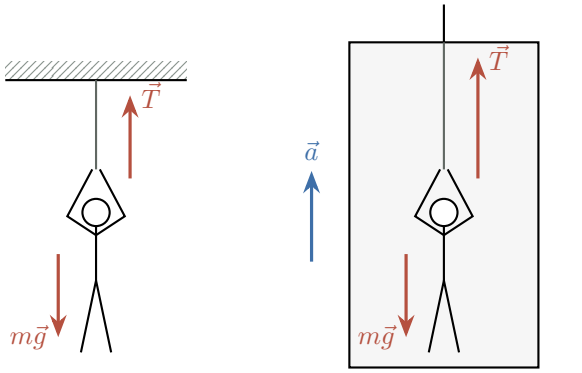
reading of the scale:

$$ma = mg - n, \qquad \Rightarrow \qquad n = m(g - a). \tag{6.3}$$

In an elevator accelerating in the $-y$ direction at $5\,\mathrm{m/s^2}$, the scale reads about half of what it reads at rest. Equations (6.2) and (6.3) are really one result: if a_y is the y component of the acceleration (positive up, negative down), then $n = m(g + a_y)$.

Now imagine that we cut the cable that moves the elevator (and that the safety brakes fail and air resistance is negligible). The elevator, the scale and the man then fall freely with acceleration $a = g$: the only force on each of them is gravity. With $a = g$, equation (6.3) gives $n = 0$: the scale reads zero, and we feel weightless. But the man's weight has not disappeared. Gravity still pulls him with the force mg, and it is exactly this force that makes him fall. He is weightless only in the sense that nothing pushes on him: his *apparent* weight is zero. **Free fall** means motion under gravity alone, and in free fall the apparent weight is always zero.

Finally, n can never be negative: a scale can push on your feet, but it cannot pull them. If the elevator were pulled down with an acceleration larger than g, equation (6.3) would give $n < 0$. What really happens is that the man leaves the scale and drifts toward the ceiling of the elevator, until the ceiling pushes him down.



(a) at rest (b) accelerating upward

Figure 6.2. Weight is misleading, again: a man hanging from a rope. (a) At rest: $T = mg$. (b) The rope hangs from the ceiling of an elevator that accelerates upward: $T = m(g + a)$.

Now let us solve the same problem for a man who holds on to a rope, as in Figure 6.2. The rope is massless, so it simply passes the pull of the ceiling on to the man. The man in the first picture is in equilibrium. Then the tension T in the rope is

$$T = mg, \tag{6.4}$$

in other words, the tension measures the weight of the man who hangs on the rope; a spring scale cut into the rope would read T. As in the previous example, let the rope hang from the ceiling of the elevator, and suppose that the elevator, and with it the man, has an acceleration a in the $+y$ direction. Then

$$ma = T - mg, \qquad \Rightarrow \qquad T = m(g + a), \tag{6.5}$$

that is, the tension in the rope increases. When the acceleration points down (with magnitude a), the tension in the rope, just as in the elevator example, is

$$ma = mg - T, \qquad \Rightarrow \qquad T = m(g - a). \tag{6.6}$$

If the rope comes loose from the ceiling, the rope and the man both start to fall freely, and there is no tension in the rope at all ($T = 0$). The tension of the rope plays exactly the role of the scale reading: it is the apparent weight of the body that hangs from it. And, like n, a tension can never be negative: a rope can pull, but it cannot push.

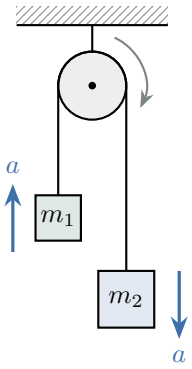


Figure 6.3. Mass and apparent weight: an Atwood machine. Two bodies, $m_2 > m_1$, hang from the ends of a massless rope that runs over an ideal pulley. Released from rest, the system turns clockwise: m_1 accelerates upward and m_2 downward, both with acceleration of magnitude a.

In the last two examples we saw that the tension in a rope and the reading of a scale tell us the same thing. Now consider two bodies hanging from the two ends of a massless rope that runs over a pulley, with no friction between the rope and the pulley, as in Figure 6.3, and let $m_2 > m_1$. (This arrangement is called an **Atwood machine**.) Released from rest, this system turns clockwise with an acceleration a: the body m_1 accelerates in the $+y$ direction and m_2 in the $-y$ direction. Because the rope does not stretch, whenever m_1 rises by some distance, m_2 goes down by the same distance, so the two bodies always have velocities, and accelerations, of the same magnitude.

One point here needs care: **because the rope is massless** and **there is no friction between the rope and the pulley**, the tension is the same at every point of the rope. To see this, suppose for a moment that the two tensions pulling up and down on a short piece of the rope, of mass Δm, are different, say T_2 and T_1. By Newton's second law this piece then accelerates with $a = (T_2 - T_1)/\Delta m$. But the rope is massless, $\Delta m = 0$, so if $T_2 \neq T_1$ this acceleration would be infinite. Put the other way round, for any finite acceleration $T_2 - T_1 = (\Delta m)\, a = 0$. So the tension must be the same at every point of a massless rope. For the piece of rope that wraps around the pulley, the same conclusion needs an *ideal* pulley: either the rope slides over it without friction, or the pulley turns freely on a frictionless axle and is itself massless. An ideal pulley changes the direction of the tension but not its magnitude. (A pulley with mass needs a torque to speed it up, and then the two tensions differ; we will meet this case when we study rotation.) So

$$T_{\text{left}} = T_{\text{right}}, \tag{6.7}$$

or, in other words, since the tension of the rope measures the apparent weight, the apparent weights of these two bodies must be the same! These two bodies, with different masses and different weights $m_1 g$ and $m_2 g$, have the same apparent weight T: a spring scale cut into the rope just above either of them would show the same reading.

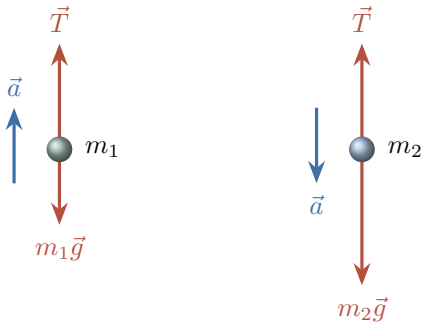


Figure 6.4. Free-body diagrams of the two bodies of the Atwood machine. The same tension T pulls up on both. For m_1 the acceleration points up, so $T > m_1 g$; for m_2 it points down, so $T < m_2 g$.

If we want to find the acceleration of the system and the tension in the rope, it helps to isolate the bodies and study them one at a time. First consider the body m_1, as in Figure 6.4. The forces on it are its weight $m_1 g$, due to gravity, and the tension T of the rope. Since the system turns clockwise, the acceleration of m_1 points up, in the same direction as T; for this body we take this direction as positive. Then Newton's second law for m_1 gives

$$m_1 a = T - m_1 g, \tag{6.8}$$

so the tension is

$$T = m_1(a + g). \tag{6.9}$$

This agrees with what we learned from the man hanging on the rope: when the acceleration points along the tension, the apparent weight of the body increases ($T > m_1 g$). For m_2 the acceleration points down, so for this body we take *down* as positive. (We are free to choose a different axis for each body, as long as we use each one consistently; choosing each axis along the body's own acceleration makes a positive for both.) Then

$$m_2 a = m_2 g - T, \tag{6.10}$$

and

$$T = m_2(g - a). \tag{6.11}$$

Equations (6.9) and (6.11) give the same tension, so $m_1(a+g) = m_2(g-a)$. Collecting the terms with a on one side, $(m_1 + m_2)\, a = (m_2 - m_1)\, g$, so the acceleration of the system is

$$a = \left(\frac{m_2 - m_1}{m_2 + m_1}\right) g. \tag{6.12}$$

Putting this into (6.9) (or (6.11)),

$$T = m_1 g \left(\frac{m_2 - m_1}{m_1 + m_2} + 1\right) = m_1 g \, \frac{(m_2 - m_1) + (m_1 + m_2)}{m_1 + m_2},$$

we find the tension in the rope:

$$T = \frac{2 m_1 m_2}{m_1 + m_2}\, g. \tag{6.13}$$

Rather than accepting these equations as dumb strings of symbols, it is important to check whether they make sense. First, the units: in (6.12) the ratio of masses has no units, so a

has the units of g, as an acceleration should; in (6.13) the fraction has the units of mass, so T has the units of mass times acceleration, that is, of force. Next, special cases. If the two masses m_1 and m_2 are equal, the system should not accelerate, and then the tension in the rope should be equal to the weight of each body. Using the acceleration (6.12) we find, as expected,

$$\lim_{m_1 \to m_2} \left(\frac{m_2 - m_1}{m_2 + m_1} \right) g = 0, \tag{6.14}$$

and in this limit the tension (6.13) becomes $T = 2m^2 g/(2m) = mg$, as expected. (Since both expressions are continuous, the limit is simply their value at $m_1 = m_2 = m$.) Next, when does the tension in the rope vanish, or in other words, when do we have free fall? Looking at the tension (6.13), we see that this happens when either of the masses is zero, which is what we expect: a body tied to a massless rope that nothing pulls on falls freely. Indeed, if we let m_1 go to zero in the acceleration (6.12), we find $a = g$: the other body, m_2, falls freely, as expected. If instead we let $m_2 \to 0$, we find $a = -g$; the minus sign tells us that now m_1 accelerates downward, falling freely. One more case is less obvious: if m_2 is much larger than m_1, then $a \to g$ but $T \to 2m_1 g$, not $m_2 g$. The heavy body falls almost freely, so the rope hardly needs to hold it back; and to give the light body an upward acceleration of almost g, the rope must pull it with almost twice its weight.

Let us now put in some numbers and test what we have learned. As we can see in Figure 6.4, the tension must be larger than the weight of m_1, $m_1 g$ (otherwise m_1 could not accelerate upward), and smaller than the weight of m_2, $m_2 g$ (otherwise m_2 could not accelerate downward). So the tension in the rope must satisfy $m_1 g < T < m_2 g$. To test this, let $m_1 = 1.1\,\mathrm{kg}$ and $m_2 = 1.25\,\mathrm{kg}$. Then equation (6.12) gives the acceleration of the system,

$$a = \frac{0.15\,\mathrm{kg}}{2.35\,\mathrm{kg}}\, g \approx 0.064\, g \approx 0.63\,\mathrm{m/s^2}, \tag{6.15}$$

and equation (6.13) gives the tension, $T = [2(1.1\,\mathrm{kg})(1.25\,\mathrm{kg})/2.35\,\mathrm{kg}]\, g \approx (1.17\,\mathrm{kg})\, g \approx 11.5\,\mathrm{N}$. As expected, this value lies between $m_1 g \approx 10.8\,\mathrm{N}$ and $m_2 g \approx 12.25\,\mathrm{N}$. Notice how small the acceleration is when the masses are nearly equal. This is exactly why George Atwood invented the machine in 1784: it "dilutes" gravity, and makes the motion slow enough to be timed with the clocks of his day, so that the laws of accelerated motion can be tested.

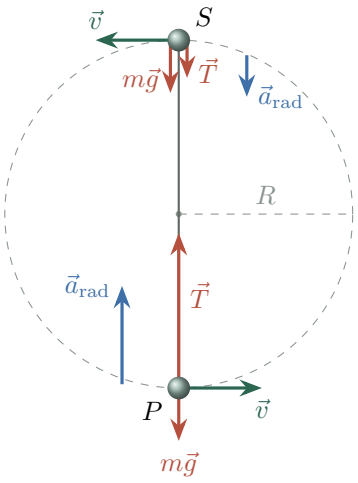


Figure 6.5. A body on a string, whirled counterclockwise in a vertical circle of radius R. At the lowest point P and the highest point S the only forces on it are the tension $\vec{T}$ of the string and its weight $m\vec{g}$ (red). Their sum must produce the centripetal acceleration $\vec{a}_{\mathrm{rad}}$, of magnitude v^2/R (blue), which points toward the center; the acceleration is not a force, and is drawn beside the body. The body moves faster at P than at S, so a_{rad} and the tension are larger at P (arrows not to scale).

Apparent weight gives interesting results when we study it in circular motion. Consider a small body tied to a string and whirled counterclockwise in a vertical circle of radius R, as in Figure 6.5. Recall from Section 4.2 that a body moving on a circle with speed v has a radial (centripetal) acceleration of magnitude $a_{\text{rad}} = v^2/R = \omega^2 R$, directed toward the center. If only gravity and the string act on the body, its speed is *not* constant: it slows down on the way up and speeds up on the way down, so this is nonuniform circular motion (Section 4.2.2), and the acceleration also has a tangential component. At the lowest point P and the highest point S, however, both forces are vertical, that is, along the radius. There is no tangential force there, so the tangential acceleration is zero and the acceleration is purely radial: $a_{\text{rad}} = v^2/R$, with v the speed at that point.

To solve the problem, at each point we choose the $+y$ axis along the centripetal acceleration, toward the center. Which forces act on the body when it is at the point P of Figure 6.5? Only two: the tension T of the string, upward, and the force of gravity, mg, downward. The centripetal acceleration is not a force, and ma_{rad} never appears in a free-body diagram: it is what the net force must produce (we come back to this in Section 6.4). Since the body continues its circular motion, Newton's second law in the radial direction (positive upward at P) requires

$$ma_{\text{rad}} = T - mg, \tag{6.16}$$

so the tension in the string is

$$T = m(a_{\text{rad}} + g). \tag{6.17}$$

This result is just like the one we found for the elevator and for the man hanging on the rope, with a replaced by a_{rad}: at P the acceleration points up, along the tension, and the apparent weight increases. If $a_{\text{rad}} = g$ at P, the apparent weight is $2mg$. When the body is at the top, at S, the center is below it, so we take the positive direction downward. Now both forces point toward the center, and the second law reads

$$ma_{\text{rad}} = T + mg, \tag{6.18}$$

so the tension in the string is

$$T = m(a_{\text{rad}} - g), \tag{6.19}$$

that is, the apparent weight decreases. If $a_{\text{rad}} = g$ at S, the tension in the string, and so the apparent weight, is zero: at that instant gravity alone supplies exactly the centripetal acceleration that the circle requires, and the body is momentarily in free fall. From (6.19) one might think that if $a_{\text{rad}} < g$ the string would have a negative tension, but this is wrong. A tension can never be negative, since a string cannot push, and this result is meaningless. It only means that the body can never reach S while moving on the circle with a taut string. If the speed it would have at the top is less than $\sqrt{gR}$, gravity alone would pull it toward the center more strongly than the circle requires; the string goes slack before the top, and the body leaves the circle and follows a projectile path inside it. (A rigid rod is different: it can push, and for a body on a rod $T < 0$ simply means that the rod pushes the body outward.) So a body on a string goes all the way round only if, at the top, $v^2 \geq gR$. Because the body is faster at the bottom, both a_{rad} and the tension are always larger at P than at S; with the energy methods of a later chapter we will find that $T_P - T_S = 6mg$, whatever the speed (as long as the string stays taut).

Example 6.1: Water in a bucket

You swing a bucket of water in a vertical circle of radius $R = 1.0\,\text{m}$. What is the smallest speed at the top for which the water stays in the bucket?

At the top, two forces act on the water: its weight mg and the push n of the bottom of the bucket, both downward, toward the center. Newton's second law in the radial direction gives $mv^2/R = n + mg$, so $n = m(v^2/R - g)$. The water stays in contact with the bucket as long as $n \geq 0$, that is, as long as $v^2 \geq gR$:

$$v_{\text{min}} = \sqrt{gR} = \sqrt{(9.8\,\text{m/s}^2)(1.0\,\text{m})} \approx 3.1\,\text{m/s}.$$

At exactly this speed the water is weightless at the top: it is in free fall, and so is the bucket, so they move together without pressing on each other. The mass of the water does not appear in the answer.

As we mentioned above, the apparent weight of a body in free fall is zero. Environments with zero apparent weight are very important for research centers such as NASA, for example for training astronauts. Such "zero-gravity" flights are now also flown commercially, for fun, for anyone who wants to experience weightlessness. As in Figure 6.6, consider an airplane that starts its maneuver with speed $v_0 = 680\,\text{km/h}$ at an angle of 45° above the horizontal. At that moment the pilots throttle back the engines, just enough that their thrust balances the air drag, and they fly the airplane so that its wings give no lift. (The engines are not switched off: then the drag would no longer be balanced.) Thrust, drag and lift then add up to zero, the net force on the airplane is its weight alone, and the airplane is a projectile. The x and y components of its initial velocity are

$$v_{0x} = v_{0y} = v_0 \cos 45° \approx 481\,\text{km/h} \approx 134\,\text{m/s}. \tag{6.20}$$

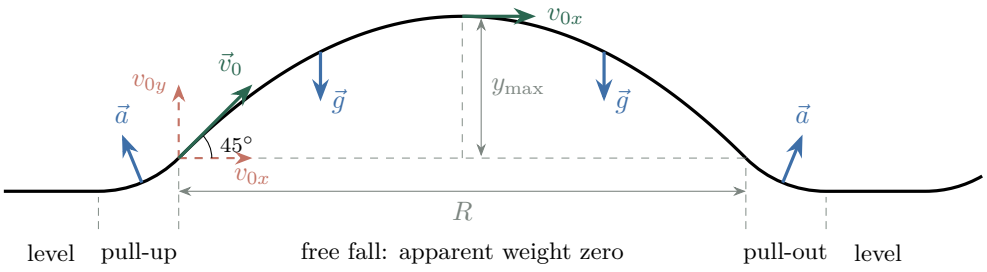


Figure 6.6. A zero-gravity flight. From level flight the airplane pulls up to 45°; then, with the thrust just balancing the drag and the wings giving no lift, it follows a parabola in free fall, and everything on board floats. In the pull-up and the pull-out its path curves upward, its acceleration $\vec{a}$ points up, and the apparent weight of everything on board is larger than its weight. The figure is not to scale.

This is exactly the projectile motion we studied in Section 4.1. The airplane starts its motion with this initial velocity, rises to a maximum height and comes back down to the level where it started; during the whole parabola, on the way up as well as on the way down, it is in free fall. From equation (4.14), with $\theta = 45°$, the maximum height it reaches

above the starting level is

$$y_{\max} = \frac{(v_0 \sin\theta)^2}{2g} \approx 910\,\mathrm{m}, \tag{6.21}$$

and from (4.13), the time it takes to reach this height and the total time of the parabola are

$$t_{\mathrm{top}} = \frac{v_0 \sin\theta}{g} \approx 13.6\,\mathrm{s} \qquad \Rightarrow \qquad t_{\mathrm{tot}} = 2t_{\mathrm{top}} \approx 27\,\mathrm{s}. \tag{6.22}$$

During this time the horizontal displacement of the airplane, the range R of Section 4.1 (not to be confused with the radius R of the circle above), is

$$R = v_{0x}\, t_{\mathrm{tot}} \approx 3.6\,\mathrm{km}. \tag{6.23}$$

During the free fall the apparent weight of the airplane and of the passengers inside it is zero: everything on board falls with the same acceleration g as the cabin, so nothing presses on anything else, and the passengers float. When the airplane gets back to the level where it started, it is diving at 45° below the horizontal at 680 km/h. The pilots now increase the thrust and pull the nose up. The airplane follows a path that curves upward, so its acceleration points up, and, as at the lowest point P of the vertical circle above, the apparent weight of the airplane and of the passengers increases (in real flights, to about 1.8 times their weight). The airplane then flies level for a while; to begin the next parabola it pulls up again, with the apparent weight again larger than the weight, until it reaches the desired initial velocity, and then the pilots throttle back once more. (Real parabolic flights give about 20 to 25 seconds of weightlessness per parabola.) The ESA videos in the box below show such flights.

Weightless astronauts. Astronauts on the International Space Station float for the same reason: the station and everything in it are in free fall around the Earth. At the station's altitude, about 400 km, the gravitational acceleration is still about 8.7 m/s^2, nearly 90 % of its value on the ground, so "zero gravity" is a misnomer: it is the *apparent* weight that is zero. The station does not hit the ground because it also moves sideways, at about 7.7 km/s; as it falls, the curved surface of the Earth falls away beneath it.

Video: *Why Are Astronauts Weightless?*[a] (Veritasium, Derek Muller, 2012, a short segment). The common answer, that there is no gravity in space, is wrong: the station is only about 400 km up. The astronauts and the station fall together, and their sideways motion keeps them from hitting the Earth.

Video: *Parabolic Flight Campaign instructions*[b] (European Space Agency, ESA/Novespace, 2024, 10 min). ESA's flight coordinator briefs the researchers whose experiments will fly, with a tour of the Novespace base in Bordeaux. Each parabola is flown as in Figure 6.6: a pull-up (apparent weight about 1.8 times the weight), about 22 s of free fall (the ideal parabola of (6.22) gives 27 s), and a pull-out.

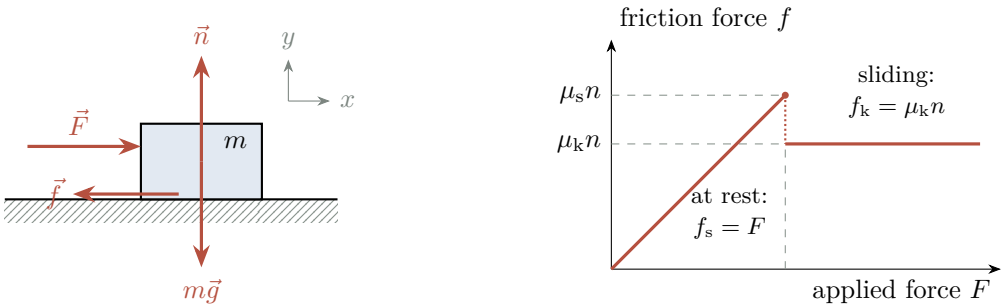
Video: *The History of Parabolic Flights*[c] (ESA, 2002, 6–7 min). Airplanes flying parabolas since the 1950s, for astronaut training and research. Watch the people on board float only during the free-fall arc, and be pressed into the floor in the pull-up and the pull-out.

[a] https://www.youtube.com/watch?v=iQOHRKKNNLQ

[b] https://www.esa.int/ESA_Multimedia/Videos/2024/09/Parabolic_Flight_Campaign_instructions

[c] https://www.esa.int/ESA_Multimedia/Videos/2002/05/The_History_of_Parabolic_Flights

6.2 Friction forces



(a) (b)

Figure 6.7. Friction. (a) A body on a horizontal surface, pushed by a horizontal force $\vec{F}$. The surface exerts on it a normal force $\vec{n}$, perpendicular to the surface, and a friction force $\vec{f}$, parallel to it. (b) The magnitude of the friction force as F is slowly increased. While the body is at rest, static friction is equal to F; it can grow only up to $\mu_s n$. Once the body slides, the friction is kinetic and drops to $\mu_k n$.

Consider a body of mass m resting on a horizontal surface, as in Figure 6.7(a). To study the problem, we choose the $+y$ axis opposite to the direction of gravity, and the $+x$ axis along the horizontal force $\vec{F}$ that we apply to the body. Since the body does not accelerate in the y direction, the surface must push up on it with a force that is equal in magnitude and opposite in direction to its weight. This is the **normal force** n ("normal" means perpendicular to the surface):

$$n = mg. \tag{6.24}$$

Two warnings. First, $n = mg$ is *not* a general rule. It holds here only because the surface is horizontal, the applied force is horizontal, and $a_y = 0$. If we pushed down on the body at an angle, or if the surface were in an accelerating elevator, n would be different, and on an incline it is smaller (equation (6.26) below). For instance, if we pull the body with a rope at an angle ϕ above the horizontal, the y equation becomes $n + F\sin\phi - mg = 0$, so $n = mg - F\sin\phi < mg$: the rope carries part of the weight, and, as we will see, the largest possible friction becomes smaller too. The normal force is whatever force the surface must exert to keep the body from sinking into it. Second, $\vec{n}$ and $m\vec{g}$ are not an action–reaction pair: both act on the same body. The third-law partner of $\vec{n}$ is the push of the body on the surface.

When we first apply the force F, the body does not necessarily start moving at once. The reason is the **friction force** between the body and the surface it rests on, which acts parallel to the surface and adjusts itself to the applied force: as long as the body stays at rest, the friction force is exactly equal and opposite to $\vec{F}$, so that the net horizontal force is zero. If we keep increasing the force, a moment comes when the body starts to move. This means that the friction force has reached its largest possible value, and F has exceeded it. The largest value the friction force on a body at rest can take is

$$f_{s,\max} = \mu_s n, \tag{6.25}$$

where μ_s is the **coefficient of static friction**. It is a ratio of two forces, so it has no units. Note that it is the *largest possible* friction force that is proportional to the *normal*

force n that presses the surfaces together, not to the weight (the two are equal only in simple situations like (6.24)); below this limit the friction is whatever the applied force requires. (The normal force and the friction force are really the perpendicular and parallel components of a single contact force that the surface exerts on the body.) For dry surfaces there are two coefficients of friction, static and kinetic:

- **Coefficient of static friction μ_s:** it describes the friction that acts while the two surfaces do not slide on each other, the friction that resists a force trying to set the body in motion. Static friction is not given by an equation but by an *inequality*: its magnitude is whatever is needed to prevent sliding, up to a maximum, $f_s \leq \mu_s n$, and its direction is opposite to the direction in which the surfaces would slide if there were no friction.
- **Coefficient of kinetic friction μ_k:** it describes the friction that acts once the surfaces slide on each other, the friction that resists the continuing motion. Its magnitude is fixed, $f_k = \mu_k n$, and to a good approximation independent of the speed; its direction is opposite to the velocity of the body relative to the surface.

$$f_s \leq \mu_s\, n \quad \text{(no sliding)}, \qquad\qquad f_k = \mu_k\, n \quad \text{(sliding)}.$$

For most pairs of surfaces $\mu_k < \mu_s$: once a body starts to slide, the friction force drops (Figure 6.7(b)). This is why it is harder to start pushing a heavy box than to keep it moving. Some typical values are listed in Table 6.1. Notice also that friction opposes the *relative sliding* of the two surfaces, not motion in general. When you walk, or when a box rides on the bed of a truck that speeds up, static friction points forward, and it is what makes you, or the box, accelerate.

Table 6.1. Approximate coefficients of friction. The values depend strongly on the condition of the surfaces (clean or dirty, dry or wet, polished or rough).

Surfaces	μ_s	μ_k
Steel on steel	0.74	0.57
Glass on glass	0.94	0.40
Rubber on dry concrete	1.0	0.8
Rubber on wet concrete	0.30	0.25
Waxed wood on wet snow	0.14	0.10
Ice on ice	0.1	0.03
Teflon on Teflon	0.04	0.04

There is more than one way to measure the coefficient of friction. First consider the inclined plane in Figure 6.8. We can resolve the gravitational force on the body of mass m on the incline into two perpendicular components. (They are not two extra forces: together they *are* $m\vec{g}$, so when we add up the forces we use either $m\vec{g}$ or its two components, never both; this is why the components are dashed in Figure 6.8.) If we choose the direction of the normal force as $+y$, one of them is $mg\cos\theta$ in the $-y$ direction, and the other is $mg\sin\theta$, down the slope, and we choose this direction as $+x$. (The angle between $m\vec{g}$ and the $-y$ axis is θ: the $-y$ axis is perpendicular to the incline and $m\vec{g}$ is perpendicular to

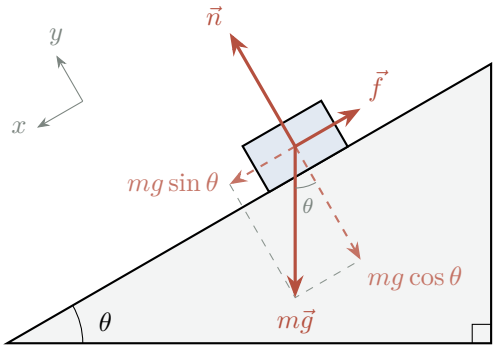


Figure 6.8. A body on an inclined plane. The weight $m\vec{g}$ (red) is resolved into a component $mg\sin\theta$ down the slope (the $+x$ direction) and a component $mg\cos\theta$ perpendicular to the surface (dashed). The angle between $m\vec{g}$ and the $-y$ direction is θ, the angle of the incline. The surface exerts the normal force $\vec{n}$ and the friction force $\vec{f}$, up the slope.

the horizontal, and two angles whose sides are mutually perpendicular are equal.) Since there is no acceleration along the y axis, the normal force must be

$$n = mg\cos\theta. \tag{6.26}$$

The body tends to move in the $+x$ direction, so friction acts in the $-x$ direction, with magnitude f_s, to prevent this motion; as long as the body stays at rest, $f_s = mg\sin\theta$. As the slope of the surface increases, that is, as the angle θ grows, the force that tends to move the body in the $+x$ direction, $mg\sin\theta$, grows, and the friction force must grow with it. At the same time the normal force $mg\cos\theta$, and with it the largest friction the surface can supply, $\mu_s mg\cos\theta$, gets smaller. So at a certain angle the pull down the slope wins over the largest possible friction, and the body starts to move. There is therefore a **critical angle** θ_c at which the body is just about to move: if the surface is tilted even slightly beyond it, the body slides, and if it is tilted slightly less, the body stays at rest. At the critical angle the static friction has just reached its maximum, $f_{s,max} = \mu_s n = \mu_s mg\cos\theta_c$, and Newton's second law along x tells us

$$mg\sin\theta_c - f_{s,max} = 0, \qquad \mu_s = \tan\theta_c, \tag{6.27}$$

since $mg\sin\theta_c = \mu_s mg\cos\theta_c$. Measuring the angle at which a block starts to slide therefore gives μ_s directly. Once the block slides, the friction is kinetic and smaller, $f_k = \mu_k mg\cos\theta$, so the block accelerates down the slope with $a = g(\sin\theta - \mu_k\cos\theta)$. If we then lower the incline until a block that has been given a push slides down with constant velocity ($a = 0$), the same argument gives $\mu_k = \tan\theta$ at that angle.

The result (6.27) is quite interesting and far from intuitive:

1. **The critical angle does not depend on the mass of the body.** Both the force that pulls the body down the slope and the largest friction force are proportional to m, so m cancels. As a result, even if you double the mass of the body, it starts to slide at the same critical angle.

2. **The friction force does not depend on the area of contact** between the body and the surface. So whether we try to slide a brick on its broad face or on its narrow side makes no difference. This does not follow from (6.27); it is an experimental fact,

built into the model (6.25), in which the maximum friction depends only on μ_s and n. The microscopic reason is that even surfaces that look smooth touch only at the tips of tiny bumps. The true area of contact is a tiny fraction of the apparent area, and it grows in proportion to the force that presses the surfaces together, not with the apparent area.

These two results, together with the fact that the largest static friction force, and the kinetic friction force, are proportional to the normal force (for the incline, (6.26)), were first found by Leonardo da Vinci, around 1500. But since Leonardo kept his findings in his notebooks, they were rediscovered by Guillaume Amontons in 1699, and are known today as Amontons' laws of friction. Keep in mind that they are approximate, empirical laws for dry surfaces: they fail, for example, for very clean and smooth metal surfaces and for lubricated surfaces, and they are only rough approximations for soft materials such as rubber.

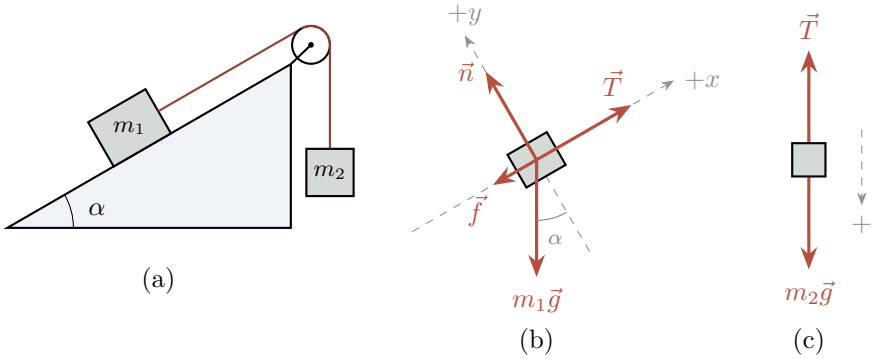


Figure 6.9. Measuring the coefficient of friction with a pulley system. (a) A body m_1 on an incline of angle α is tied by a rope, over an ideal pulley, to a hanging body m_2. (b) Free-body diagram of m_1, with the $+x$ axis up the slope; the friction $\vec{f}$ is drawn for a body that tends to move in the $+x$ direction (case 1). (c) Free-body diagram of m_2, with the positive direction downward.

To measure the coefficient of friction, now consider a pulley system like the one in Figure 6.9. Let m_1 be the mass of the body on the inclined surface and m_2 the mass of the hanging body. We can resolve the forces on the body on the surface into components as we did in the previous problem, but this time we take the $+x$ axis up the slope, along the rope. Here, however, the situation is radically different from the previous problem: we do not know in advance whether the system will accelerate in the $+x$ direction (m_1 up the slope and m_2 down), in the $-x$ direction, or stay at rest. As we will see shortly, these three cases have to be treated separately, and no single equation describes all of them, because static friction is given by an inequality and its direction depends on which way the body tends to slide.

If we draw the free-body diagrams of the bodies, as on the right of Figure 6.9, we first see that the normal force is again $n = m_1 g \cos\alpha$. The only open question in the analysis of the system is which way the friction points: if the body tends to move in the $+x$ direction, friction acts in the $-x$ direction, but if the body tends to move in the $-x$ direction, friction acts in the $+x$ direction. All we know about the static friction force is that its maximum value is

$$f_{s,\max} = \mu_s n = \mu_s m_1 g \cos\alpha. \tag{6.28}$$

The last thing we know about the system is that the rope is massless and the pulley

frictionless, so the tension is the same at every point of the rope (Section 6.1). So while the system is at rest, Newton's second law for the hanging body (with $a = 0$) gives the tension in the rope:

$$T = m_2 g. \tag{6.29}$$

With this information we can study the three cases we mentioned, for a system that starts from rest.

1. **The system is just about to accelerate in the $+x$ direction.** In this limiting case the friction force has its maximum magnitude and points in the $-x$ direction. So Newton's second law for the body m_1, which is still at rest, reads

$$T - m_1 g \sin\alpha - f_{\mathrm{s,max}} = 0, \qquad \Rightarrow \qquad m_2 g = m_1 g(\sin\alpha + \mu_\mathrm{s}\cos\alpha). \tag{6.30}$$

   This is the critical situation in which the body is held in equilibrium by the largest possible friction force. If m_2 becomes even a hair heavier, the system starts to accelerate in the $+x$ direction. The condition for this is

$$m_2 g > m_1 g(\sin\alpha + \mu_\mathrm{s}\cos\alpha). \tag{6.31}$$

2. **The system is just about to accelerate in the $-x$ direction.** In this limiting case the friction force has its maximum magnitude and points in the $+x$ direction. So Newton's second law for the body m_1 reads

$$T - m_1 g \sin\alpha + f_{\mathrm{s,max}} = 0, \qquad \Rightarrow \qquad m_2 g = m_1 g(\sin\alpha - \mu_\mathrm{s}\cos\alpha). \tag{6.32}$$

   This again is the critical situation in which the body is held in equilibrium by the largest possible friction force. If m_2 becomes even a hair lighter, the system starts to accelerate in the $-x$ direction, with m_1 sliding down the slope. The condition for this is

$$m_2 g < m_1 g(\sin\alpha - \mu_\mathrm{s}\cos\alpha). \tag{6.33}$$

   If $\tan\alpha \le \mu_\mathrm{s}$, the right-hand side is zero or negative, and this case can never happen: even with no hanging body at all, m_1 would not slide down by itself (compare (6.27)).

3. **The system stays at rest.** If neither (6.31) nor (6.33) holds, that is, if $m_1 g(\sin\alpha - \mu_\mathrm{s}\cos\alpha) \le m_2 g \le m_1 g(\sin\alpha + \mu_\mathrm{s}\cos\alpha)$, the acceleration is zero, and the static friction adjusts itself to keep the system in equilibrium. Its x component is then $f_x = m_1 g\sin\alpha - m_2 g$, which may be positive (up the slope) or negative (down the slope), and its magnitude is at most $f_{\mathrm{s,max}}$.

Example 6.2: Will it move?

As a very simple example, take $m_1 = 1\,\mathrm{kg}$, $m_2 = 2\,\mathrm{kg}$, $\alpha = 30°$, a coefficient of static friction $\mu_\mathrm{s} = 0.5$ and a coefficient of kinetic friction $\mu_\mathrm{k} = 0.4$, and let the system start from rest. To understand how the system behaves, we must check the two conditions above. To keep the arithmetic simple, in this example we take $g = 10\,\mathrm{m/s^2}$. Then

$$m_2 g = 20\,\mathrm{N}, \qquad m_1 g\sin\alpha = 5\,\mathrm{N}, \qquad f_{\mathrm{s,max}} = \mu_\mathrm{s} m_1 g\cos\alpha \approx 4.33\,\mathrm{N}. \tag{6.34}$$

So $m_2 g = 20\,\mathrm{N}$ is larger than $m_1 g\sin\alpha + f_{\mathrm{s,max}} \approx 9.33\,\mathrm{N}$: the first condition holds,

and the system accelerates in the $+x$ direction.

Since the first condition holds and the system starts to move, we must now use the coefficient of kinetic friction, not the static one, to learn more about it. The body m_1, sliding up the slope, feels a kinetic friction force down the slope, of magnitude

$$f_\mathrm{k} = \mu_\mathrm{k} n = \mu_\mathrm{k} m_1 g \cos\alpha. \tag{6.35}$$

Applying Newton's second law to the body m_1 along the x axis, we find

$$T - m_1 g \sin\alpha - \mu_\mathrm{k} m_1 g \cos\alpha = m_1 a. \tag{6.36}$$

For the body m_2, with the positive direction downward, along its acceleration, Newton's second law reads $m_2 a = m_2 g - T$, so

$$T = m_2(g - a). \tag{6.37}$$

(Note that now $T \neq m_2 g$: equation (6.29) holds only while the system is at rest.) Putting this value of T into (6.36) and solving for a,

$$a = \frac{m_2 g - m_1 g \sin\alpha - \mu_\mathrm{k} m_1 g \cos\alpha}{m_1 + m_2} = \frac{20\,\mathrm{N} - 5\,\mathrm{N} - 3.46\,\mathrm{N}}{3\,\mathrm{kg}},$$

and then using this acceleration in (6.37), we find the acceleration and the tension in the rope:

$$a \approx 3.85\,\mathrm{m/s^2}, \qquad T \approx 12.3\,\mathrm{N}. \tag{6.38}$$

The tension T, which measures the apparent weight of the body m_2, has come out smaller than the weight of m_2, 20 N, as expected, since m_2 accelerates downward.

Now let us keep everything else in this example the same, but change the mass m_2 to $m_2 = 0.4\,\mathrm{kg}$. Then $m_2 g = 4\,\mathrm{N}$, which satisfies neither the first condition ($4\,\mathrm{N} > 9.33\,\mathrm{N}$ is false) nor the second ($4\,\mathrm{N} < 5\,\mathrm{N} - 4.33\,\mathrm{N} = 0.67\,\mathrm{N}$ is false). So for $m_2 = 0.4\,\mathrm{kg}$ the acceleration is zero: the friction force adjusts itself to keep the system in equilibrium. With $a = 0$ the tension is $T = m_2 g$, and, with f_x the x component of the friction force, the second law for m_1 along x reads

$$\underbrace{-m_1 g \sin\alpha}_{5\,\mathrm{N}} + \underbrace{T}_{m_2 g = 4\,\mathrm{N}} + f_x = 0, \qquad f_x = 1\,\mathrm{N}. \tag{6.39}$$

The friction force points in the $+x$ direction, up the slope, and has a magnitude of 1 N, well below its maximum of 4.33 N, as it must be. Notice that it is *not* $\mu_\mathrm{s} n$: static friction is only as large as it needs to be.

Video: *Lecture 4 – Newton's Laws (cont.) and Inclined Planes*[a] (Open Yale Courses, Ramamurti Shankar, 2006, about 67 min; an optional full lecture, with a transcript). Friction, a block on an incline, pulleys (Section 6.1) and friction in circular motion (Section 6.4). Watch for static friction as an inequality, $f_\mathrm{s} \leq \mu_\mathrm{s} n$, as in "Will it move?", and compare his incline with (6.27).

[a] https://oyc.yale.edu/physics/phys-200/lecture-4

6.3 Fluid resistance and terminal speed

In the previous section the friction between dry surfaces was, to a good approximation, independent of the area of contact and of the speed of the body; its largest static value, and the kinetic friction, depended on the body only through the normal force. The **fluid resistance** we discuss in this section, the force that a fluid (a gas or a liquid, such as air or water) exerts on a body moving through it, does depend on these quantities: on the speed of the body relative to the fluid, and on the size and shape of the body. A simple model that works well in many situations gives the magnitude of this force as

$$f = kv + Dv^2, \tag{6.40}$$

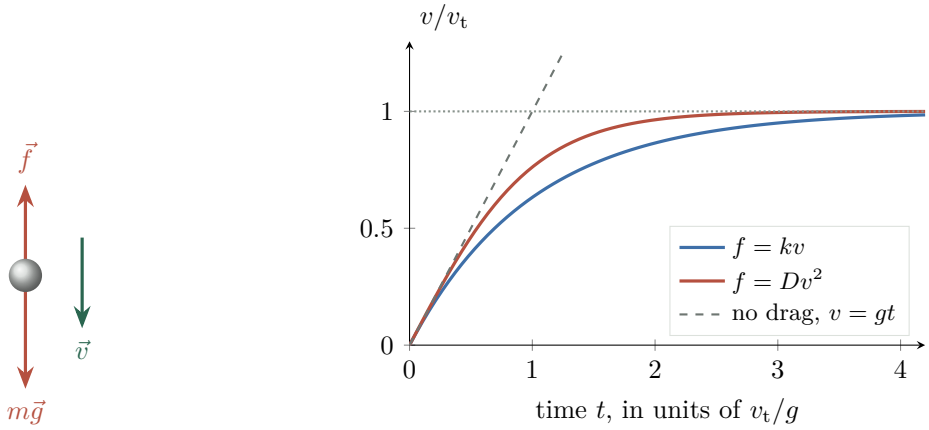
where v is the speed of the body relative to the fluid, and the force points opposite to the velocity of the body relative to the fluid: $\vec{f} = -(k + Dv)\,\vec{v}$. The constants k (units kg/s) and D (units kg/m) depend on the fluid and on the size and shape of the body. The first term in equation (6.40) is the "viscous" term, and the second the "pressure" term. **Viscosity** is a measure of how strongly a fluid resists flowing, that is, resists a shear deformation, in which neighboring layers of the fluid slide past each other; in everyday words, it describes how "thick" or "sticky" the fluid is. If we think of water, oil and tar, we see at once that these three liquids have different viscosities. If the viscosity of the fluid is very large, k is very large. (For a sphere of radius r moving slowly through a fluid of viscosity η, $k = 6\pi\eta r$; this is Stokes' law.) Viscosity depends strongly on temperature: if you heat tar, it becomes liquid and deforms much more easily, while if you cool it, it becomes solid and very hard to deform. The coefficient D of the second term is related to the density ρ of the fluid. A moving body has to push the fluid in front of it out of its way, and this takes a force that grows with the density of the fluid and with the cross-sectional area A of the body: in each second the body sweeps through a volume Av of fluid, of mass ρAv, and sets it moving with a speed of the order of v; the momentum it gives the fluid each second, which is the force (Section 5.3), is therefore of order ρAv^2. This is why the second term grows as v^2, and why $D = \frac{1}{2}C\rho A$, where the drag coefficient C depends on the shape of the body (it is about 0.5 for a sphere).

Now consider a body dropped from rest in a fluid, as in Figure 6.10(a); we take the positive direction downward. (This is not free fall, since gravity is not the only force.) At first $v = 0$ and there is no fluid resistance, so the body starts with acceleration g. As it speeds up, the resistance grows, as we see from equation (6.40), so the net force $mg - f$, and with it the acceleration, decreases. The speed approaches the value at which the two forces would be equal; as it does, the net force, and with it the acceleration, approaches zero, and the body moves with an almost constant velocity. This speed is called the **terminal speed** v_t:

$$mg = kv_\mathrm{t} + Dv_\mathrm{t}^2. \tag{6.41}$$

(Here we neglect the buoyant force of the fluid, which is a good approximation in air; in a liquid we replace mg by the weight minus the buoyant force.) Strictly speaking, the body gets closer and closer to v_t but never quite reaches it; Figure 6.10(b) shows how fast it gets close. Equation (6.41) is a quadratic equation, and it gives two solutions for the terminal speed,

$$v_\mathrm{t} = \frac{-k \pm \sqrt{k^2 + 4Dmg}}{2D}.$$



(a) (b)

Figure 6.10. Terminal speed. (a) Free-body diagram of a body falling through a fluid: its weight $m\vec{g}$ and the fluid resistance $\vec{f}$, opposite to the velocity $\vec{v}$. (b) The speed of a body dropped from rest, for a resistance $f = kv$ (blue) and $f = Dv^2$ (red). Both curves start with the slope of free fall (dashed) and approach the terminal speed v_t (dotted) without ever quite reaching it.

The product of the two roots is $-mg/D < 0$, so one root is positive and the other negative. Only the positive root corresponds to a physical situation; the negative one is not a possible speed, and we discard it.

To see which property of the fluid contributes more to the resistance, and how the terminal speed changes in these different cases, consider first the speed at which the viscous term and the pressure term exert equal forces. We call this speed the **critical speed** v_c:

$$kv_c = Dv_c^2, \qquad \Rightarrow \qquad v_c = \frac{k}{D}. \tag{6.42}$$

Which kind of resistance dominates depends on how the speed of the body compares with the critical speed. If we write the magnitude of the resistance as

$$f = k\Big(v + \frac{D}{k}v^2\Big) = kv\Big(1 + \frac{v}{v_c}\Big), \tag{6.43}$$

we see that there are two cases that we can study easily.

1. $v \ll v_c$: the fluid resistance comes mostly from viscosity ($f \approx kv$), and the terminal speed is

   $$mg = kv_t, \qquad \Rightarrow \qquad v_t = \frac{mg}{k}. \tag{6.44}$$

   This is the case for very small or slow bodies, or for bodies in very viscous liquids: dust and fog droplets in air, or a steel ball sinking in honey. (The approximation is consistent only if the v_t we find is indeed much smaller than v_c.)

2. $v \gg v_c$: the resistance comes mostly from the pressure term, whose coefficient is proportional to the density of the fluid ($f \approx Dv^2$), and the terminal speed is

   $$mg = Dv_t^2, \qquad \Rightarrow \qquad v_t = \sqrt{\frac{mg}{D}}. \tag{6.45}$$

   This is the case for almost everything that moves fast through air: balls, raindrops, skydivers, cars. (Again, the approximation is consistent only if the v_t we find is much larger than v_c.)

How quickly is the terminal speed approached? For the viscous law, Newton's second law reads $m\,\mathrm{d}v/\mathrm{d}t = mg - kv = k(v_\mathrm{t} - v)$. You can check, by differentiating, that the solution that starts from rest is

$$v(t) = v_\mathrm{t}(1 - e^{-kt/m}).$$

After a time $m/k = v_\mathrm{t}/g$ the body has reached 63 % of its terminal speed, and after $4.6\,m/k$ it is within 1 % of it. For the quadratic law a similar calculation gives $v(t) = v_\mathrm{t}\tanh(gt/v_\mathrm{t})$. Both are plotted in Figure 6.10(b). Both curves start with slope g: at the very beginning, when the body hardly moves and the resistance is still negligible, it is falling almost freely.

Example 6.3: A skydiver and a cloud droplet

For a human body falling through air with arms and legs spread out, $D \approx 0.25\,\mathrm{kg/m}$. A skydiver of mass 80 kg (with equipment) moves much faster than v_c, so (6.45) gives

$$v_\mathrm{t} = \sqrt{\frac{mg}{D}} = \sqrt{\frac{(80\,\mathrm{kg})(9.8\,\mathrm{m/s^2})}{0.25\,\mathrm{kg/m}}} = 56\,\mathrm{m/s} \approx 200\,\mathrm{km/h}.$$

Here $v_\mathrm{t}/g \approx 5.7\,\mathrm{s}$, and since $\tanh 2.65 \approx 0.99$, she is within 1 % of this speed about $2.65 \times 5.7\,\mathrm{s} \approx 15\,\mathrm{s}$ after she jumps. Since $v_\mathrm{t} \propto \sqrt{m}$, a heavier body of the same size and shape falls faster through air.

A cloud droplet of radius $r = 10\,\mu\mathrm{m}$ is in the opposite, viscous regime. With Stokes' law, $k = 6\pi\eta r$, and the viscosity of air, $\eta \approx 1.8 \times 10^{-5}\,\mathrm{kg/(m\,s)}$, we find $k \approx 3.4 \times 10^{-9}\,\mathrm{kg/s}$. The droplet's mass is $m = \frac{4}{3}\pi r^3 \rho_\mathrm{water} = \frac{4}{3}\pi(1.0\times10^{-5}\,\mathrm{m})^3(1000\,\mathrm{kg/m^3}) \approx 4.2 \times 10^{-12}\,\mathrm{kg}$, so (6.44) gives $v_\mathrm{t} = mg/k \approx 1.2 \times 10^{-2}\,\mathrm{m/s}$, about 1 cm/s. (Check: with $D = \frac{1}{2}C\rho A$, $C \approx 0.5$ and $\rho_\mathrm{air} \approx 1.2\,\mathrm{kg/m^3}$, the critical speed is $v_\mathrm{c} = k/D \approx 36\,\mathrm{m/s}$, far above v_t, so the viscous approximation is consistent.) The slightest updraft keeps such droplets up, and this is why clouds float.

In a vacuum there is no fluid resistance at all, and a feather falls exactly like a bowling ball. Watch the experiment in NASA's vacuum chamber: *feathers and a bowling ball in a vacuum chamber*.[a]

[a] https://www.youtube.com/watch?v=E43-CfukEgs

6.4 Dynamics of circular motion

In Section 4.2.1 we saw that a particle in uniform circular motion, moving with speed v on a circle of radius R, has an acceleration of magnitude $a_\mathrm{rad} = v^2/R$ that points toward the center. By Newton's second law, the net force on the particle must then also point toward the center, with magnitude

$$\sum F_\mathrm{rad} = ma_\mathrm{rad} = m\,\frac{v^2}{R} \qquad \text{(net force toward the center)}.$$

This net force is often called the **centripetal force**, but it is not a new kind of force: it is whatever combination of the forces we already know (the tension of a string, gravity, a normal force, friction) happens to point toward the center. In a free-body diagram we

draw only these real forces, and never add an extra "centripetal force" mv^2/R. Nor do we draw a "centrifugal force" pointing outward: in an inertial frame there is no such force. When a car turns left and you feel pushed to the right, nothing pushes you outward. Your body tends to keep moving in a straight line (Newton's first law), and the door has to push you inward to make you turn with the car. If the motion is not uniform, the same equation holds for the radial component of the net force, with v the speed at that instant, as we used for the vertical circle in Section 6.1; the tangential component of the net force changes the speed.

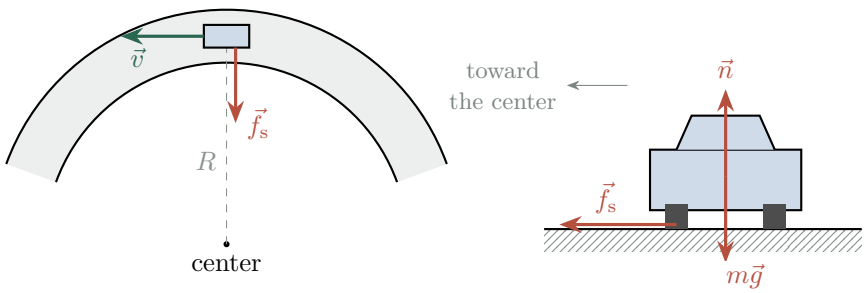
Example 6.4: A car on a flat curve

A car of mass m takes a flat (unbanked) curve of radius R at constant speed v (Figure 6.11). What supplies the centripetal force, and how fast can the car go?

Vertically, nothing accelerates, so $n = mg$. Horizontally, the only force toward the center is the friction of the road on the tires. It is *static* friction, even though the car is moving: the tires roll without slipping, so the part of each tire that touches the road is momentarily at rest relative to the road. So $f_s = mv^2/R$, and since $f_s \le \mu_s n = \mu_s mg$,

$$m\frac{v^2}{R} \le \mu_s mg, \qquad \Rightarrow \qquad v \le \sqrt{\mu_s g R}.$$

For $R = 50\,\text{m}$ and rubber on dry concrete ($\mu_s = 1.0$, Table 6.1), this gives $v_{\max} \approx 22\,\text{m/s} \approx 80\,\text{km/h}$; on wet concrete ($\mu_s = 0.30$) it drops to about $12\,\text{m/s}$, or $44\,\text{km/h}$. The mass cancels: on the same curve, with the same tires, a heavy truck and a light car have the same maximum speed. Above it, the tires slip and the car slides outward: not because something pushes it out, but because friction can no longer bend its path into the circle.



(a) seen from above (b) seen from behind

Figure 6.11. A car on a flat curve. The car turns left, so seen from behind the center of the curve is on its left. The only horizontal force on the car, the static friction $\vec{f}_s$ of the road on the tires, points toward the center of the curve and supplies the centripetal force mv^2/R.

Video: *11.2 Worked Example – Car on a Banked Turn*[a] (MIT OpenCourseWare, 8.01SC, 2016, short). The same car on an icy, frictionless road banked at an angle ϕ: the normal force supplies the centripetal force, and $n\cos\phi = mg$, $n\sin\phi = mv^2/R$ give $v = \sqrt{gR\tan\phi}$, the only speed at which the car can take the curve. As on the flat road, the mass cancels.

[a] https://ocw.mit.edu/courses/8-01sc-classical-mechanics-fall-2016/pages/week-3-circular-motion/11-2-worked-example-car-on-a-banked-turn/

Videos and simulations for this week

- *Why Are Astronauts Weightless?* (Veritasium): Section 6.1, weightless astronauts.
- *Parabolic Flight Campaign instructions* (ESA): Section 6.1, zero-gravity flights.
- *The History of Parabolic Flights* (ESA): Section 6.1, zero-gravity flights.
- *Lecture 4 – Newton's Laws (cont.) and Inclined Planes* (Open Yale Courses, Ramamurti Shankar; optional full lecture): Section 6.2, friction and inclines.
- *Feathers and a bowling ball in NASA's vacuum chamber* (BBC, *Human Universe*, Brian Cox): Section 6.3, falling with no air resistance.
- *11.2 Worked Example – Car on a Banked Turn* (MIT OpenCourseWare, 8.01SC): Section 6.4, the banked curve.

About these notes

These notes are not original. They draw heavily on the books listed in the Week 1 notes, on Walter Lewin's 8.01x Classical Mechanics lectures at MIT, and on the Dynamics and Relativity courses taught at different times by Stephen Siklos and David Tong at the University of Cambridge. They are not official course notes of FIZ 101: they are written only to help the students of Mehmet Özkan's section. Istanbul Technical University and its Department of Physics Engineering bear no responsibility for them.